

Data Flow Diagrams (DFD) & Agile User Stories

Project: Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer

Document Reference: REQ-PHASE2-DFD-US-V1.0

Target Environment: ServiceNow Utah / Vancouver / Washington DC / Xanadu LTS

System Module: Service Catalog, Flow Designer, ITAM (alm_hardware)

1. Executive Summary

This deliverable provides the comprehensive data architectural model and functional specification for the automated Standard Laptop Procurement system. It articulates the precise information exchanges, transformations, database interactions, and user requirements driving the automated workflow.

The document is organized into three major sections:

- Data Flow Architecture:** High-fidelity Level 0 Context Diagram and Level 1 Functional Decomposition Diagram rendered in Mermaid format.
- ServiceNow Core Data Dictionary:** Schema specifications for all primary transactional, governance, asset, and foundational tables.
- Agile Epics & User Stories:** 8 detailed user stories structured across 3 epics totaling 26 story points, complete with MoSCoW prioritization and Gherkin acceptance criteria (Given-When-Then).

2. Data Flow Architecture

2.1. DFD Level 0: Enterprise Context Diagram

The Level 0 Context Diagram establishes the operational system boundary. It defines the four external human and systemic entities interacting with the **0.0 Laptop Procurement Automation System** (powered by ServiceNow Flow Designer) and illustrates all inbound and outbound data streams.

```
graph TD
    classDef entity fill:#1E293B,stroke:#3B82F6,stroke-width:2px,color:#FFFFFF;
    classDef system fill:#0F172A,stroke:#10B981,stroke-width:3px,color:#FFFFFF;

    REQ[👤 Requester / Employee]:::entity
    MGR[👤 Approving Line Manager]:::entity
    TECH[🔧 Hardware Staging Engineer]:::entity
    ASSET[📦 IT Asset Management / ITAM]:::entity

    SYS((⚡ 0.0 Laptop Procurement<br/>Automation System<br/>ServiceNow Flow Designer)):::system

    REQ -->|1. Inbound: Laptop Order Submission<br/>Hardware Tier, Delivery Method, Justification| SYS
    SYS -->|2. Outbound: Real-time Order Tracking<br/>Stage Progression, Delivery ETA, CSAT|
```

Survey | REQ

SYS --> | 3. Outbound: Actionable Approval Request
Employee Profile, Unit Cost, Hardware Tier | MGR

MGR --> | 4. Inbound: Approval Decision
Approved / Rejected + Mandatory Comments | SYS

SYS --> | 5. Outbound: Auto-Generated Catalog Task
Configuration Specs, Delivery Address, SLA Clock | TECH

TECH --> | 6. Inbound: Staging Completion Payload
Asset Tag Barcode, Serial Number, MAC Address | SYS

SYS --> | 7. Outbound: Lifecycle State & Ownership Update
Status: In Use, Assigned To: Requester | ASSET

ASSET --> | 8. Inbound: Stock Availability & Depot Inventory Data | SYS

Inbound & Outbound Data Stream Catalog (Level 0)

Stream ID	Direction	External Entity	Payload Description	Triggering Event / Frequency
DS-01	Inbound	Requester / Employee	Catalog submission parameters: <code>requested_for</code> , <code>hardware_bundle</code> , <code>delivery_method</code> , <code>shipping_address</code> , <code>business_justification</code> .	On-demand submission via Service Portal (<code>/esc</code>).
DS-02	Outbound	Requester / Employee	Real-time stage progression updates (<code>waiting_for_approval</code> , <code>fulfillment</code> , <code>delivery</code> , <code>complete</code>), shipment tracking number, first-login setup guide, and post-delivery CSAT survey.	Event-driven on flow stage changes and task closure.
DS-03	Outbound	Approving Line Manager	Actionable approval payload: Requester name, department, cost center, hardware specifications, total unit cost, justification text, and cryptographic 1-click decision tokens.	Flow action <code>Ask for Approval</code> on RITM insert.
DS-04	Inbound	Approving Line Manager	Approval decision token (<code>approved</code> or <code>rejected</code>), digital signature timestamp, and required rejection rationale comments.	Manager action via Outlook Actionable Message, ServiceNow Mobile, or Portal.
DS-05	Outbound	Hardware Staging Engineer	Pre-populated fulfillment task (<code>sc_task</code>): Hardware bundle specifications, OS build version, delivery destination, SLA deadline (24 hours), and parent RITM link.	Automatic generation upon successful approval.
DS-06	Inbound	Hardware Staging Engineer	Physical staging completion payload: Scanned <code>u_asset_tag</code> , verified manufacturer serial number, network MAC address, and checklist completion confirmation.	Technician marks <code>sc_task</code> as Closed Complete.
DS-07	Outbound	IT Asset Management (ITAM)	Relational database write updating <code>alm_hardware</code> : Transition <code>install_status</code> from <code>6</code> (In Stock) to <code>1</code> (In Use), clear <code>substatus</code> , set <code>assigned_to</code> = Requester GUID, update <code>location</code> .	Transactional commit on <code>sc_task</code> closure.

DS-08	Inbound	IT Asset Management (ITAM)	Hardware catalog metadata, model product numbers (<code>cmdb_hardware_product_model</code>), active depot stock levels, and available asset tag pools.	Real-time query during form rendering and asset allocation.
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2.2. DFD Level 1: Functional Process Decomposition Diagram

The Level 1 Diagram decomposes the system into five discrete automated and human processing stages, detailing their bidirectional interactions with ServiceNow core database stores.

```
graph TD
    classDef proc fill:#1E1E2E,stroke:#89B4FA,stroke-width:2px,color:#CDD6F4;
    classDef store fill:#181825,stroke:#F9E2AF,stroke-width:2px,color:#CDD6F4;
    classDef entity fill:#11111B,stroke:#A6E3A1,stroke-width:2px,color:#CDD6F4;

    %% External Entities
    E_REQ[👤 Requester / Employee]:::entity
    E_MGR[👤 Line Manager]:::entity
    E_TECH[🔧 Hardware Staging Tech]:::entity

    %% Discrete Functional Processes
    P1[1.0 Catalog Submission<br/>& Variable Validation]:::proc
    P2[2.0 Approval Routing<br/>& Governance Engine]:::proc
    P3[3.0 Automated Task<br/>Provisioning & Dispatch]:::proc
    P4[4.0 Hardware Staging,<br/>Imaging & Asset Binding]:::proc
    P5[5.0 Delivery Verification<br/>& Lifecycle Closure]:::proc

    %% Database Stores (ITSM / ITAM Tables)
    D1[(D1: sc_request<br/>Parent Order Container)]:::store
    D2[(D2: sc_req_item<br/>Requested Item Context)]:::store
    D3[(D3: sysapproval_approver<br/>Governance Approval Log)]:::store
    D4[(D4: sc_task<br/>Catalog Fulfillment Task)]:::store
    D5[(D5: alm_hardware<br/>Enterprise Hardware Assets)]:::store
    D6[(D6: sys_user<br/>Identity & Org Hierarchy)]:::store

    %% Process 1.0 Data Flows
    E_REQ -->|Submit Model, Site, Justification| P1
    D6 -. ->|Query User Profile, Department & Manager GUID| P1
    P1 -->|Insert Parent Order Record| D1
    P1 -->|Insert RITM with Bound Variables| D2

    %% Process 2.0 Data Flows
    D2 -->|Database Insert Trigger: State=Open| P2
    P2 -->|Resolve Approver Hierarchy| D6
    P2 -->|Generate Approval Record: State=Requested| D3
    E_MGR <-->|Dispatch Notification / Ingest Actionable Response| P2
    P2 -->|Commit Decision: State=Approved/Rejected| D3
    P2 -->|Update RITM: Approval=Approved, Stage=Fulfillment| D2

    %% Process 3.0 Data Flows
    P2 -->|Signal Flow Execution on Approval| P3
    P3 -->|Insert Pre-Configured Task Record| D4
    P3 -->|Map RITM Variables to Task Variable Editor| D4

    %% Process 4.0 Data Flows
```

```

D4 -->|Present in Depot Queue with 24h OLA| E_TECH
E_TECH -->|Scan Barcode / Query Stock Availability| D5
E_TECH -->|Submit Asset Tag, Serial, Image Status| P4
P4 -->|Validate Tag & Transition State: In Stock -> Reserved| D5
P4 -->|Commit Task Closure: State=Closed Complete| D4

%% Process 5.0 Data Flows
D4 -->|Trigger Flow on Task State = Closed Complete| P5
P5 -->|Update Asset: install_status=In Use, assigned_to=Requester| D5
P5 -->|Update RITM: State=Closed Complete, Stage=Complete| D2
P5 -->|Cascade Close Parent Request: request_state=closed_complete| D1
P5 -->|Send Delivery Confirmation & CSAT Survey| E_REQ

```

Detailed Process Specifications (Level 1)

• Process 1.0: Catalog Submission & Variable Validation

- *Input Data:* User input selections from Service Portal form; session context from `g_user`.
- *Transformation Logic:* Validates mandatory fields; evaluates client-side catalog UI policies; queries `sys_user` for requester's manager, department, and location.
- *Output Data:* Atomically creates parent `sc_request` and child `sc_req_item` with bound variable dictionary.

• Process 2.0: Approval Routing & Governance Engine

- *Input Data:* Newly inserted `sc_req_item` record; `requested_for.manager` pointer.
- *Transformation Logic:* Executes Flow Designer `Ask for Approval` action; generates actionable email with cryptographic tokens; initiates 24h reminder and 48h escalation timers; evaluates branch (Approved vs. Rejected).
- *Output Data:* Updated `sysapproval_approver` record; updated `sc_req_item.approval` (`approved/rejected`) and `stage` (`fulfillment/request_cancelled`).

• Process 3.0: Automated Task Provisioning & Dispatch

- *Input Data:* Approval signal from Process 2.0; RITM variable dictionary.
- *Transformation Logic:* Executes Flow Designer `Create Catalog Task` action; binds hardware model, delivery method, and address into `sc_task.description`; assigns to `Hardware Fulfillment Depot` group; initiates 24h staging OLA clock.
- *Output Data:* Inserted `sc_task` record in `Open` state.

• Process 4.0: Hardware Staging, Imaging & Asset Binding

- *Input Data:* Depot technician inputs (scanned asset tag barcode, serial number, imaging status).
- *Transformation Logic:* Data Policy verifies asset tag existence and `install_status == 6` (In Stock) in `alm_hardware`; technician executes automated PXE OS flash; marks task `Closed Complete`.
- *Output Data:* `sc_task.state` updated to `3` (`Closed Complete`); asset temporarily marked `Reserved`.

• Process 5.0: Delivery Verification & Lifecycle Closure

- *Input Data:* Completion event of child `sc_task`.
- *Transformation Logic:* Flow Designer detects task completion; writes final updates to `alm_hardware` (`install_status = 1 - In Use, assigned_to = sc_req_item.requested_for`); updates `sc_req_item.state = 3` (`Closed Complete`); triggers parent `sc_request` cascading closure; dispatches delivery email and 2h delayed CSAT survey.
- *Output Data:* Fully closed request container; reconciled asset record; CSAT survey record.

3. ServiceNow Core Data Dictionary & Schema Specifications

The automated procurement solution strictly utilizes standard, out-of-the-box ServiceNow ITSM and ITAM tables. No custom tables are introduced, preserving seamless platform upgradeability across LTS releases.

3.1. Request Table (`sc_request`)

Extends `task`. Acts as the commercial order container for shopping cart checkouts, billing, and high-level tracking.

Field Name	Column Label	Type	Length / Ref	Mand.	Choice Values / System Rules	Description & Automation Behavior
<code>sys_id</code>	Sys ID	<code>sys_id</code>	32 hex	Yes	System generated GUID	Primary key.
<code>number</code>	Number	<code>string</code>	40	Yes	Read-only; Prefix <code>REQ</code>	Unique auto-generated request number (e.g., <code>REQ0010042</code>).
<code>requested_for</code>	Requested For	<code>reference</code>	<code>sys_user</code>	Yes	Foreign key to <code>sys_user</code>	Employee who will receive and utilize the ordered hardware.
<code>opened_by</code>	Opened By	<code>reference</code>	<code>sys_user</code>	Yes	Defaults to current session	User who submitted the catalog item.
<code>approval</code>	Approval	<code>choice</code>	40	Yes	<code>not_requested</code> , <code>requested</code> , <code>approved</code> , <code>rejected</code>	Aggregated approval status across child requested items.
<code>request_state</code>	Request State	<code>choice</code>	40	Yes	<code>in_process</code> , <code>closed_complete</code> , <code>closed_cancelled</code>	State of overall request container. Closed automatically when all child RITMs close.
<code>price</code>	Total Price	<code>currency</code>	Decimal(15,2)	No	Calculated sum	Aggregate monetary cost of all ordered items in cart.

opened_at	Opened	glide_date_time	N/A	Yes	Auto-stamped	Timestamp marking requisition entry into the platform.
closed_at	Closed	glide_date_time	N/A	No	Auto-stamped	Timestamp set when parent request reaches terminal state.

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3.2. Requested Item Table (sc_req_item)

Extends `task`. Represents the specific standard laptop ordered and serves as the operational execution context for Flow Designer.

Field Name	Column Label	Type	Length / Ref	Mand.	Choice Values / System Rules
sys_id	Sys ID	sys_id	32 hex	Yes	System generated GUID
number	Number	string	40	Yes	Read-only; Prefix <code>RITM</code>
request	Parent Request	reference	sc_request	Yes	Foreign key
cat_item	Catalog Item	reference	sc_cat_item	Yes	Foreign key
state	State	choice	40	Yes	1 (Open), 2 (Work in Progress), 3 (Closed Complete), 4 (Closed Incomplete), 7 (Closed Skipped)

stage	Stage	choice	40	Yes	request_approved, waiting_for_approval, fulfillment, delivery, complete
approval	Approval	choice	40	Yes	not_requested, requested, approved, rejected
configuration_item	Configuration Item	reference	cldb_ci_computer	No	Foreign key
context	Flow Context	reference	sys_flow_context	System	Foreign key
work_notes	Work Notes	journal_input	4000	No	Audit journal

3.3. Catalog Task Table (sc_task)

Extends task. Represents the physical fulfillment work unit assigned to the Hardware Depot engineering team.

Field Name	Column Label	Type	Length / Ref	Mand.	Choice Values / System Rules	Description / Auto Behavior
sys_id	Sys ID	sys_id	32 hex	Yes	System generated GUID	Primary key
number	Number	string	40	Yes	Read-only; Prefix TASK	Unique generated task number (e.g., TASK1234567890)
request_item	Requested Item	reference	sc_req_item	Yes	Foreign key	Points to parent RITM

assignment_group	Assignment Group	reference	sys_user_group	Yes	Foreign key	Popu Flow to Ha Fulf Depc
assigned_to	Assigned To	reference	sys_user	No	Foreign key	Indivi techn who c the ta the d queu
short_description	Short Description	string	160	Yes	Dynamic string	E.g., Deve Stan (Win for Chen
description	Description	string	4000	Yes	Dynamic multi-line text	Form instru conta mode delive meth addre
state	State	choice	40	Yes	1 (Open), 2 (Work in Progress), 3 (Closed Complete), 4 (Closed Incomplete)	Oper stagi Moni Wait Conc flow s
priority	Priority	choice	40	Yes	1 (Critical), 2 (High), 3 (Moderate), 4 (Low)	Defau (Mod escal (High reque
u_asset_tag	Asset Tag	string	40	Yes*	Must match alm_hardware.asset_tag	*Enfc mand Data prior state Clos Comp

3.4. Hardware Asset Table (alm_hardware)

Extends `alm_asset`. Represents physical enterprise hardware inventory and tracking records.

state	State	choice	40	Yes	not_requested, requested, approved, rejected, cancelled	Approval status. Updated via Actionable Message or Portal.
comments	Comments	journal_input	4000	Cond.	Mandatory if rejected	Explanatory rationale provided by approver.
iteration	Iteration	integer	N/A	System	Default 1	Tracks approval loops in multi-tier workflows.

3.6. User & Organization Tables (sys_user & cmn_department)

Foundation tables providing identity, organizational hierarchy, and reporting structures.

Table Name	Field Name	Type	Reference / Rule	Description & Usage
sys_user	sys_id	sys_id	GUID	Unique user primary key.
sys_user	user_name	string(40)	Unique Login ID	Corporate SSO username (e.g., alex.chen).
sys_user	email	string(100)	Valid Email	Corporate email address for actionable notifications.
sys_user	manager	reference	sys_user	Direct line manager; resolved by Flow Designer approval engine.
sys_user	department	reference	cmn_department	Department hierarchy used for financial cost-center mapping.
sys_user	vip	boolean	true/false	Flag triggering expedited priority handling in Flow Designer.
cmn_department	dept_head	reference	sys_user	Department head; fallback escalation approver if manager is inactive.

4. Agile Epics & User Stories

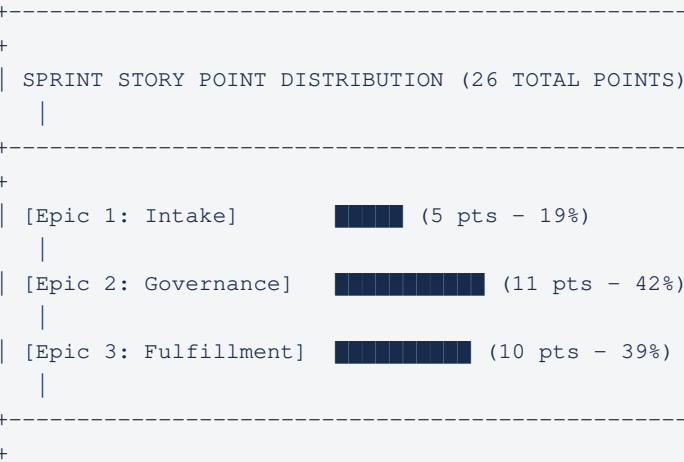
Epic Structure Overview

- **Epic 1: Self-Service Catalog & Intake Experience** (5 Story Points)
 - Focuses on simplifying discovery, enforcing mandatory input validation, and auto-populating user profile metadata.
- **Epic 2: Automated Governance, Approval & Notifications** (11 Story Points)
 - Focuses on two-tier manager approvals, actionable email integration, 24h reminders, and rejection handling.

• **Epic 3: Automated Fulfillment, Asset Lifecycle & Delivery** (10 Story Points)

- Focuses on autonomous task generation, mandatory asset tag binding, cascading record closure, and CSAT dispatch.

• **Total Story Points: 26 Story Points**



4.1. Epic 1: Self-Service Catalog & Intake Experience

User Story US-01: Intuitive Catalog Item Selection

- **Story ID:** US-01
- **Epic:** Epic 1: Self-Service Catalog & Intake Experience
- **MoSCoW Priority:** Must Have
- **Story Points:** 3 Points
- **User Story Statement:**

As an employee requiring a standard computer,

I want to select a pre-configured laptop model from the Service Catalog,

So that I do not have to guess hardware specifications or submit unstructured free-text tickets.

• **Acceptance Criteria (Gherkin):**

```
Feature: Standard Laptop Catalog Selection
  As an employee
  I want to browse standard laptop models
  So that I can request the right equipment for my job role

  Scenario: User browses available standard laptop models
    Given I am logged into the Employee Center portal (/esc) as an active employee
    When I navigate to "Hardware" > "Computers" > "Standard Laptop Order"
    Then I should see three pre-approved hardware tiers:
      | Bundle Name | Screen Size | CPU / RAM / Storage |
```

Default OS			
	Standard Business (Windows)	14-inch	Intel Core i7 / 16GB / 512GB SSD
Windows 11 Ent			
	Developer Standard (Windows)	16-inch	Intel Core i9 / 32GB / 1TB SSD
Windows 11 Ent			
	Developer Standard (macOS)	16-inch	Apple M3 Pro / 36GB / 1TB SSD
macOS Sonoma			

And each model must display its standard delivery SLA (72 business hours).

Scenario: Validation prevents submission without delivery method

Given I am on the "Standard Laptop Order" catalog form

When I select a hardware bundle but leave "Delivery Method" unselected

And I click the "Order Now" button

Then the system prevents form submission

And displays a field validation error: "Delivery Method is mandatory."

- **Technical Implementation Notes:** Defined in `sc_cat_item` with three variable choices under `hardware_bundle` Client script enforces mandatory checks prior to checkout.

User Story US-02: Automatic Profile & Manager Resolution

- **Story ID:** US-02
- **Epic:** Epic 1: Self-Service Catalog & Intake Experience
- **MoSCoW Priority:** Must Have
- **Story Points:** 2 Points
- **User Story Statement:**

As a requester,

I want my department, office location, and manager to be pre-populated on the request form,

So that I save time and eliminate data entry errors.
- **Acceptance Criteria (Gherkin):**

Feature: Automatic User Profile Resolution
As a requester
I want my corporate profile details auto-populated
So that I don't have to manually lookup my manager or cost center
Scenario: Form initializes with logged-in user profile
Given I open the "Standard Laptop Order" catalog form
When the form completes client-side loading
Then the "Requested For" field defaults to my current user account
And "Department", "Office Site", and "Line Manager" are dynamically populated from my sys_user record
And the "Line Manager" field is locked as read-only to prevent unauthorized tampering.
Scenario: Manager submits request on behalf of a direct report
Given I have the permission to order on behalf of others

When I change the "Requested For" field to team member "Alex Chen"
Then the form dynamically clears and updates "Department", "Office Site", and "Line Manager" to match "Alex Chen"
And the "Line Manager" field reflects Alex's direct manager.

- **Technical Implementation Notes:** Catalog Client Script `onLoad` and `onChange(requested_for)` executes `g_scratchpad` or GlideAjax query to resolve user record attributes cleanly without synchronous freezes.

4.2. Epic 2: Automated Governance, Approval & Notifications

User Story US-03: Automated Line-Manager Approval Routing

- **Story ID:** US-03
- **Epic:** Epic 2: Automated Governance, Approval & Notifications
- **MoSCoW Priority:** Must Have
- **Story Points:** 5 Points
- **User Story Statement:**

As a governance officer,

I want standard laptop requests to be automatically routed to the requester's direct manager via Flow Designer,

So that unapproved hardware expenditure is prevented and a strict audit trail is maintained.

- **Acceptance Criteria (Gherkin):**

Feature: Automated Line-Manager Approval Routing

As an IT governance officer

I want Flow Designer to route approval requests to direct managers

So that all hardware procurements are authorized

Scenario: Successful creation and notification of approval record

Given a new standard laptop order is submitted creating record "RITM0010085"

When Flow Designer triggers on RITM insert

Then an approval record is inserted into sysapproval_approver with:

Field	Value
sysapproval	Pointer to RITM0010085
approver	Pointer to requester's sys_user.manager
state	requested

And RITM0010085 stage updates to "waiting_for_approval"

And an actionable approval email containing hardware tier, price, and justification is sent to the manager.

Scenario: Manager approves via actionable email

Given an approval record exists in state "requested"

When the manager clicks "Approve" within the actionable email client

Then the sysapproval_approver state transitions to "approved"

And Flow Designer advances execution to the fulfillment branch

And RITM0010085 approval field updates to "approved".

- **Technical Implementation Notes:** Uses Flow Designer core action `Ask for Approval` bound to data pill `Trigger -> Requested Item Record -> Requested For -> Manager`.

User Story US-04: Approval Rejection & Request Termination

- **Story ID:** US-04
- **Epic:** Epic 2: Automated Governance, Approval & Notifications
- **MoSCoW Priority:** Must Have
- **Story Points:** 3 Points
- **User Story Statement:**

As a line manager,

I want to reject a laptop request with mandatory explanatory comments,

So that unwarranted requests are terminated immediately and the requester is informed of the business reason.

- **Acceptance Criteria (Gherkin):**

Feature: Approval Rejection Handling

As an approving line manager

I want to reject unjustified laptop requests

So that departmental budget is protected

Scenario: Manager rejects request with justification comments

Given a pending approval record exists in sysapproval_approver for RITM0010085

When the manager selects "Reject" and provides comment "Existing laptop is only 14 months old; refresh denied."

Then sysapproval_approver state transitions to "rejected"

And Flow Designer updates RITM0010085 state to "Closed Incomplete" (State = 4)

And updates RITM stage to "Request Cancelled"

And logs the verbatim rejection comments in RITM work notes

And sends a notification email to the requester containing the manager's rejection reason

And halts all downstream fulfillment task creation.

- **Technical Implementation Notes:** Flow Designer conditional `If Approval State == Rejected` branch executes `Update Record (RITM)` and `Send Email` actions before terminating the flow.

User Story US-05: Automated Approval Escalation & Reminder

- **Story ID:** US-05
- **Epic:** Epic 2: Automated Governance, Approval & Notifications
- **MoSCoW Priority:** Should Have
- **Story Points:** 3 Points
- **User Story Statement:**

As an IT service owner,

I want automated reminder notifications sent to managers who have not acted on an approval within 24 hours,

So that orders do not stagnate in manager inboxes.

- **Acceptance Criteria (Gherkin):**

Feature: Approval Reminder & Escalation

As an IT service owner

I want automated reminders sent to approvers

So that procurement lead times remain under 72 hours

Scenario: Automated reminder dispatched after 24 hours

Given an approval record for RITM0010085 has remained in state "requested" for 24 business hours

When the Flow Designer timer action fires

Then a reminder email with subject "[REMINDER] Action Required: Laptop Approval for Alex Chen" is sent to the manager

And a work note is added to RITM0010085: "24-hour approval reminder notification dispatched to manager."

Scenario: Approval escalates to Department Head at 48 hours

Given an approval has remained in state "requested" for 48 business hours

When the escalation timer fires

Then Flow Designer creates an additional approval record for the Department Head (cmn_department.dept_head)

And dispatches an urgent escalation notification.

- **Technical Implementation Notes:** Implemented via Flow Designer **Wait for Condition / Timer** action or Flow Designer approval timeout subflow.

4.3. Epic 3: Automated Fulfillment, Asset Lifecycle & Delivery

User Story US-06: Automated Catalog Task (sc_task) Creation

- **Story ID:** US-06
- **Epic:** Epic 3: Automated Fulfillment, Asset Lifecycle & Delivery
- **MoSCoW Priority:** Must Have
- **Story Points:** 3 Points
- **User Story Statement:**

As a hardware technician,

I want a pre-populated fulfillment task generated in my group queue immediately upon manager approval,

So that my team can begin staging without manual ticket triage or data re-entry.

• **Acceptance Criteria (Gherkin):**

Feature: Automated Catalog Task Generation
As a hardware depot technician
I want fulfillment tasks generated automatically upon approval
So that I can immediately begin staging hardware

Scenario: Flow Designer instantiates staging task upon approval
Given RITM0010085 transitions to approval state "approved"
When Flow Designer enters the fulfillment branch
Then an sc_task record is created within 10 seconds with:

Field	Value
request_item	RITM0010085
assignment_group	Hardware Fulfillment Depot
priority	3 (Moderate)
short_description	Deploy Developer Standard (Windows) for Alex Chen
state	1 (Open)

And the task Variable Editor displays all submitted catalog variables
And RITM0010085 stage updates to "fulfillment".

- **Technical Implementation Notes:** Flow Designer action `Create Catalog Task` binds data pills from trigger RITM and sets assignment group.

User Story US-07: Mandatory Asset Tag Binding on Task Completion

- **Story ID:** US-07
- **Epic:** Epic 3: Automated Fulfillment, Asset Lifecycle & Delivery
- **MoSCoW Priority:** Must Have
- **Story Points:** 5 Points
- **User Story Statement:**

As an IT asset manager,

I want technicians to be required to enter a valid in-stock Asset Tag before closing an `sc_task`,

So that physical hardware is never deployed untracked and asset records reflect real-time allocation.

• **Acceptance Criteria (Gherkin):**

Feature: Mandatory Asset Tag Binding

As an IT asset manager
I want mandatory asset validation before task closure
So that 100% of deployed laptops are tracked in ITAM

Scenario: Task closure aborted when Asset Tag is missing
Given an active staging `sc_task` for RITM0010085
When the technician attempts to change state to "Closed Complete" with "`u_asset_tag`" empty
Then the system blocks record save
And displays an error: "Asset Tag is mandatory to close this task."

Scenario: Successful asset reconciliation on task completion
Given an active staging `sc_task` for RITM0010085
When the technician enters a valid in-stock asset tag "AST-009482"
And sets state to "Closed Complete"
Then the task saves successfully
And the linked record in `alm_hardware` is updated:

Field	Value
<code>install_status</code>	1 (In Use)
<code>substatus</code>	NULL
<code>assigned_to</code>	Pointer to RITM requested_for user
<code>location</code>	Pointer to RITM delivery location

And an audit line is created in `sys_audit` verifying asset transfer.

- **Technical Implementation Notes:** Enforced via ServiceNow Data Policy on `sc_task` combined with Flow Designer `Update Record` action targeting `alm_hardware`.

User Story US-08: Order Stage Transparency & Cascading Closure

- **Story ID:** US-08
- **Epic:** Epic 3: Automated Fulfillment, Asset Lifecycle & Delivery
- **MoSCoW Priority:** Should Have
- **Story Points:** 2 Points
- **User Story Statement:**

As a requester,

I want the Service Portal to show my request progressing through stages and close automatically when delivered,

So that I always know the status of my laptop and receive delivery instructions.

- **Acceptance Criteria (Gherkin):**

Feature: Cascading Closure and Stage Transparency
As an employee requester
I want real-time portal tracking and delivery closure
So that I have full visibility of my order status

Scenario: All staging tasks complete and order closes cleanly
Given all child `sc_task` records for RITM0010085 are in state "Closed Complete"

When Flow Designer executes the final stage actions
Then RITM0010085 state transitions to "Closed Complete" (State = 3)
And RITM0010085 stage transitions to "complete"
And parent request REQ0010042 transitions to "closed_complete"
And an email notification with asset tag AST-009482, serial number, and onboarding setup guide is sent to the requester
And an automated CSAT survey trigger is queued for dispatch within 2 hours.

- **Technical Implementation Notes:** Flow Designer `Wait for Condition` action unblocks upon `sc_task` closure, executes `Update Record (RITM)` and `Update Record (REQ)`, and fires event `sc_req_item.laptop_delivered`

5. Requirements Traceability Matrix (User Stories vs System Components)

User Story ID	MoSCoW	Points	Target Tables	Primary Flow Designer Action / System Component	Upstream Dependency
US-01	Must	3	<code>sc_cat_item</code> , <code>item_option_new</code>	Service Catalog Item & Variable Sets	Catalog taxonomy setup
US-02	Must	2	<code>sys_user</code> , <code>cmn_department</code>	Catalog Client Script (<code>onLoad</code> , <code>onChange</code>)	Corporate LDAP / SSO sync
US-03	Must	5	<code>sc_req_item</code> , <code>sysapproval_approver</code>	Flow Action: <code>Ask for Approval</code>	US-01, US-02
US-04	Must	3	<code>sc_req_item</code> , <code>sysapproval_approver</code>	Flow Action: <code>Update Record</code> + <code>Send Email</code>	US-03
US-05	Should	3	<code>sysapproval_approver</code> , <code>sysevent</code>	Flow Action: <code>Wait for Condition</code> / Timer Subflow	US-03
US-06	Must	3	<code>sc_task</code> , <code>sc_req_item</code>	Flow Action: <code>Create Catalog Task</code>	US-03 (Approval)
US-07	Must	5	<code>sc_task</code> , <code>alm_hardware</code>	Data Policy + Flow Action: <code>Update Record (alm_hardware)</code>	US-06
US-08	Should	2	<code>sc_req_item</code> , <code>sc_request</code> , <code>asmt_metric_type</code>	Flow Action: <code>Update Record</code> + Cascading Close Engine	US-06, US-07

End of DFD and User Stories Deliverable — REQ-PHASE2-DFD-US-V1.0